

A Living Research Record: Version-Controlled Experiment Trees with Frozen Nodes

Every experiment is a node; the paper is the projection

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1 Abstract

This living record tracks $\{\{\text{EXP_TOTAL}\}\}$ experiments organized as a version-controlled tree with frozen nodes. Of these, $\{\{\text{EXP_ANSWERED}\}\}$ are answered ($\{\{\text{EXP_WINS}\}\}$ wins, $\{\{\text{EXP_DEAD_ENDS}\}\}$ dead ends), $\{\{\text{EXP_FROZEN}\}\}$ are frozen preregistrations, and $\{\{\text{EXP_PROVISIONAL}\}\}$ are provisional plans. The paper you are reading is generated from the tree state; answering or freezing a node regenerates it.

2 Introduction

Research logs rot because outcomes live in notebooks while plans live in issue trackers. We propose the opposite coupling: a single experiment tree where nodes are preregistered (frozen) before being run, and the manuscript is a deterministic projection of that tree. The tree carries a `run_command` contract per node, so every recorded experiment names the canonical command that produced its evidence.

3 Methodology

Each node carries a status from the ladder {provisional, frozen, answered}. A provisional node may be frozen (preregistered, immutable) or answered directly. Answering a frozen node is refused by construction: preregistered commitments may not be back-filled. Within one tree, each `run_command` may be declared by at most one node, so an experiment identity maps one-to-one onto the command that runs it.

4 Results

4.1 Answered nodes

{{EXP_SECTIONS_RESULTS}}

4.2 Winners per round

{{EXP_WINNERS_PER_ROUND}}

5 Dead ends and discussion

Negative results are first-class: every answered node with outcome `dead_end` is registered below, keeping future rounds away from retired directions.

{{EXP_SECTIONS_DEAD_ENDS}}

5.1 In progress (frozen preregistrations)

{{EXP_SECTIONS_IN_PROGRESS}}

6 References

See `references.bib` for the bibliography (rendered by Pandoc).